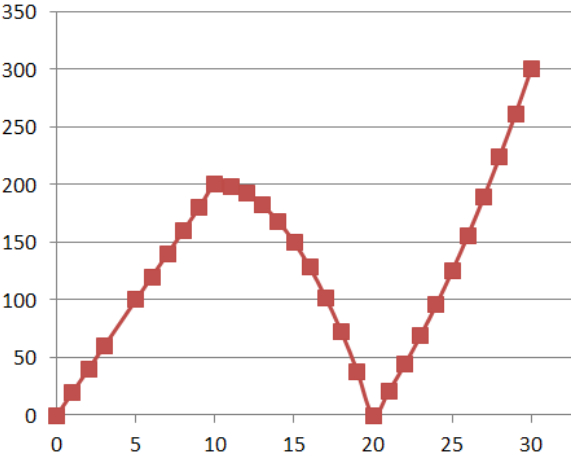


1a	Assumption: g is constant as h is very small compared to the radius of the earth. Work done by external agent (W_e) in moving a mass upward goes to the gain in GPE of the object Hence, GPE gain = Work Done = Force by external agent $\times$ Distance moved = mgh	M1 M1 A0
1b	 Rate of work done = $P = Fv$. Correct shape and value for each of the segment For 0 m s^{-1} to 10 m s^{-1} For 10 m s^{-1} to 20 m s^{-1} For 20 m s^{-1} to 30 m s^{-1}	B1 B1 B1
2ai	From the graph, at $4.0 \times 10^{-8} \text{ m}$, $g = 0.8 \text{ N kg}^{-1}$	